

Support Vector Machine

- A Support Vector Machine (SVM) is a very **powerful** and **versatile** Machine Learning model, capable of performing linear or nonlinear **classification**, **regression**, and even **outlier** detection.
- SVMs are particularly well suited for **classification of complex** but **small-** or **medium-**sized datasets.

Some Important Terms Related to SVM

Hyperplane: It is a decision boundary that separates data points of different classes in the feature space. Mathematical representation of hyperplane is $y = wx + b$

Support Vectors: These are the data points that are closest to the hyperplane. These points influence the orientation and position of the hyperplane.

Margin: It is the distance between the hyperplane and the nearest data point of either class. SVM tries to maximize this margin to improve classification confidence.

Some Important Terms Related to SVM

Hard Margin: Assume data is perfectly separable by the hyperplane. All points must lie outside the margin.

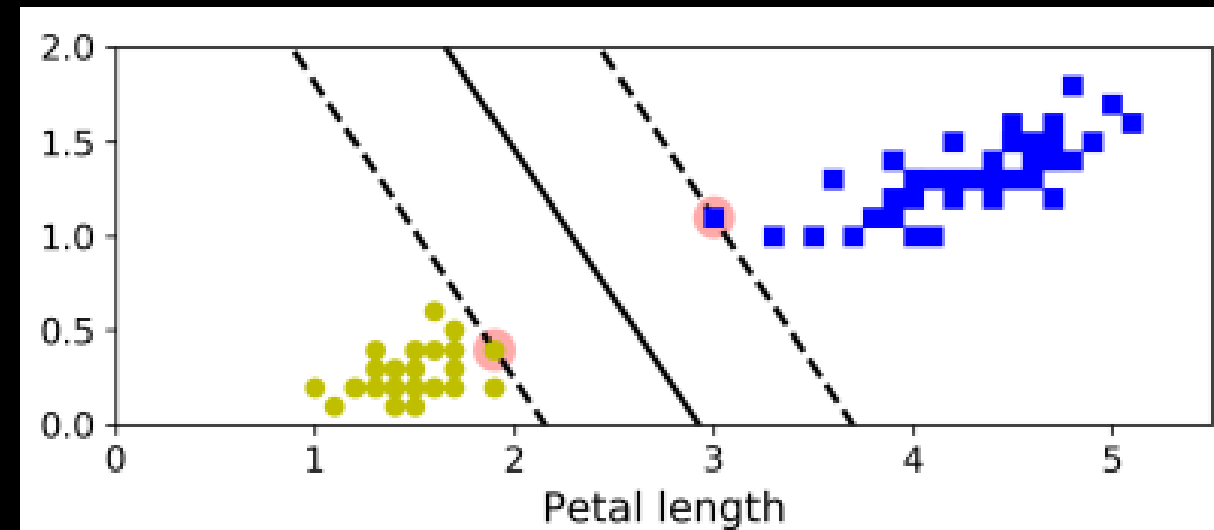
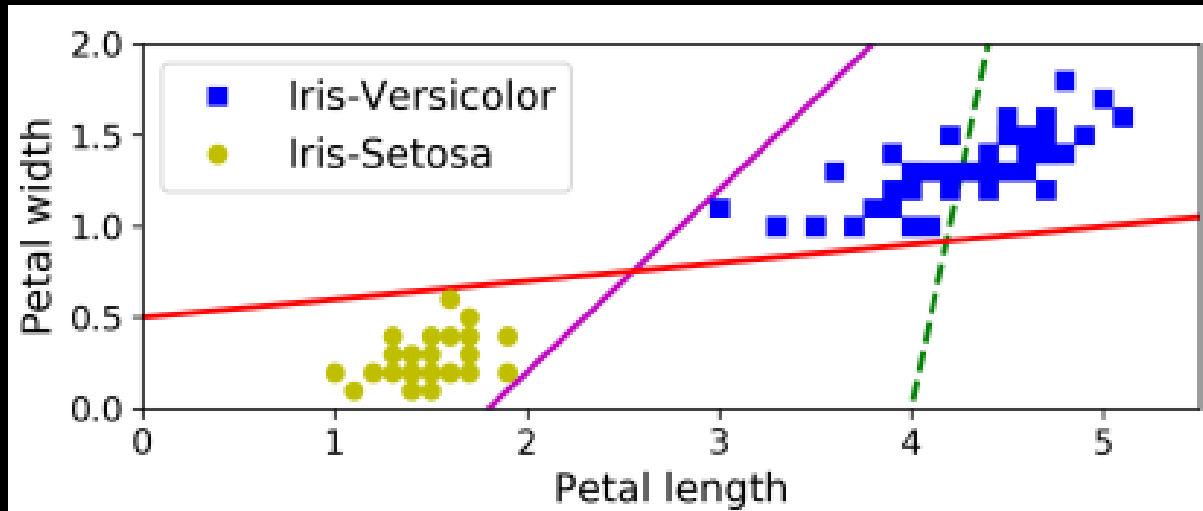
Soft Margin: Allows some misclassification or margin violations for non-linearly separable data.

Kernel Function: It transforms data into a higher dimensional space to make it linearly separable.

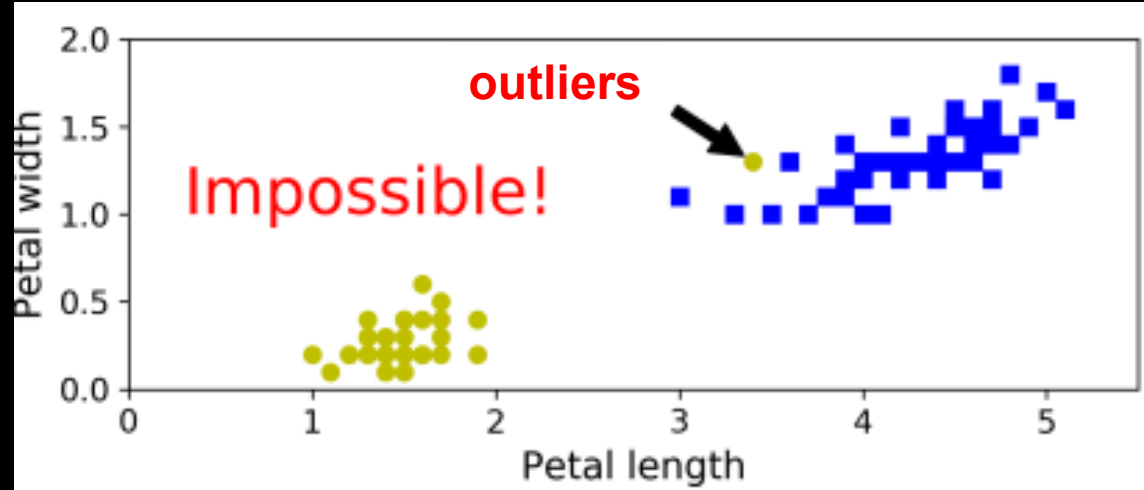
Types of Kernels:

- Linear Kernel: Suitable for linearly separable data.
- Polynomial Kernel: For curved boundaries.
- Radial Basis Function (RBF) Kernel: Captures complex relationships.

Support Vector Machine

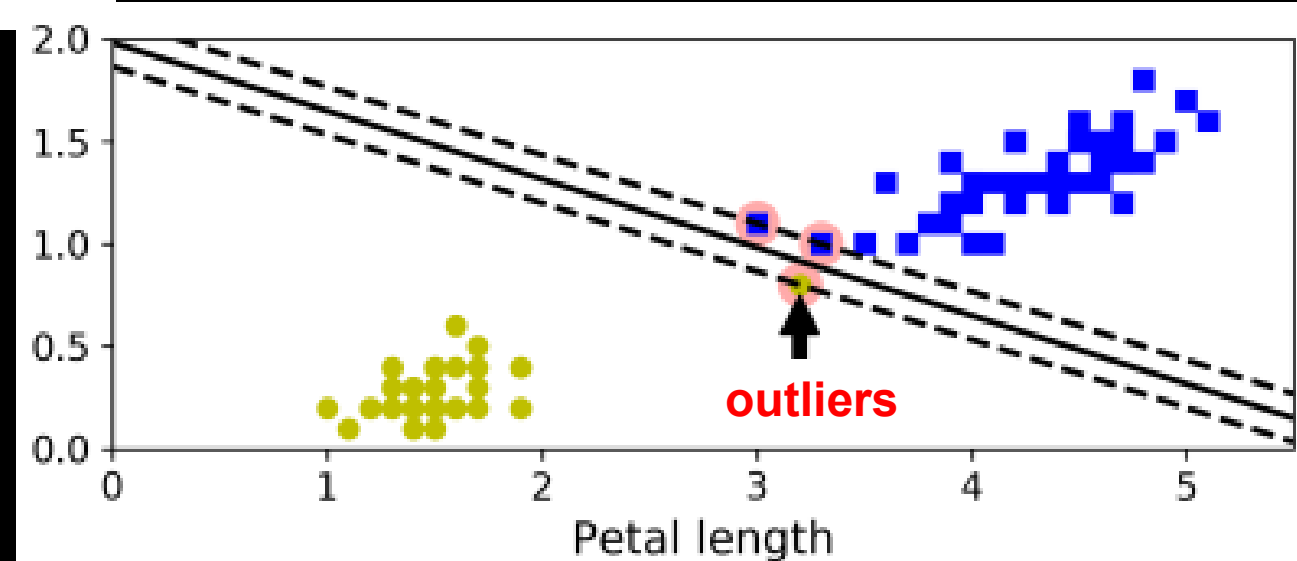


Soft Margin Classification



To avoid these issues it is preferable to use a more **flexible model**.

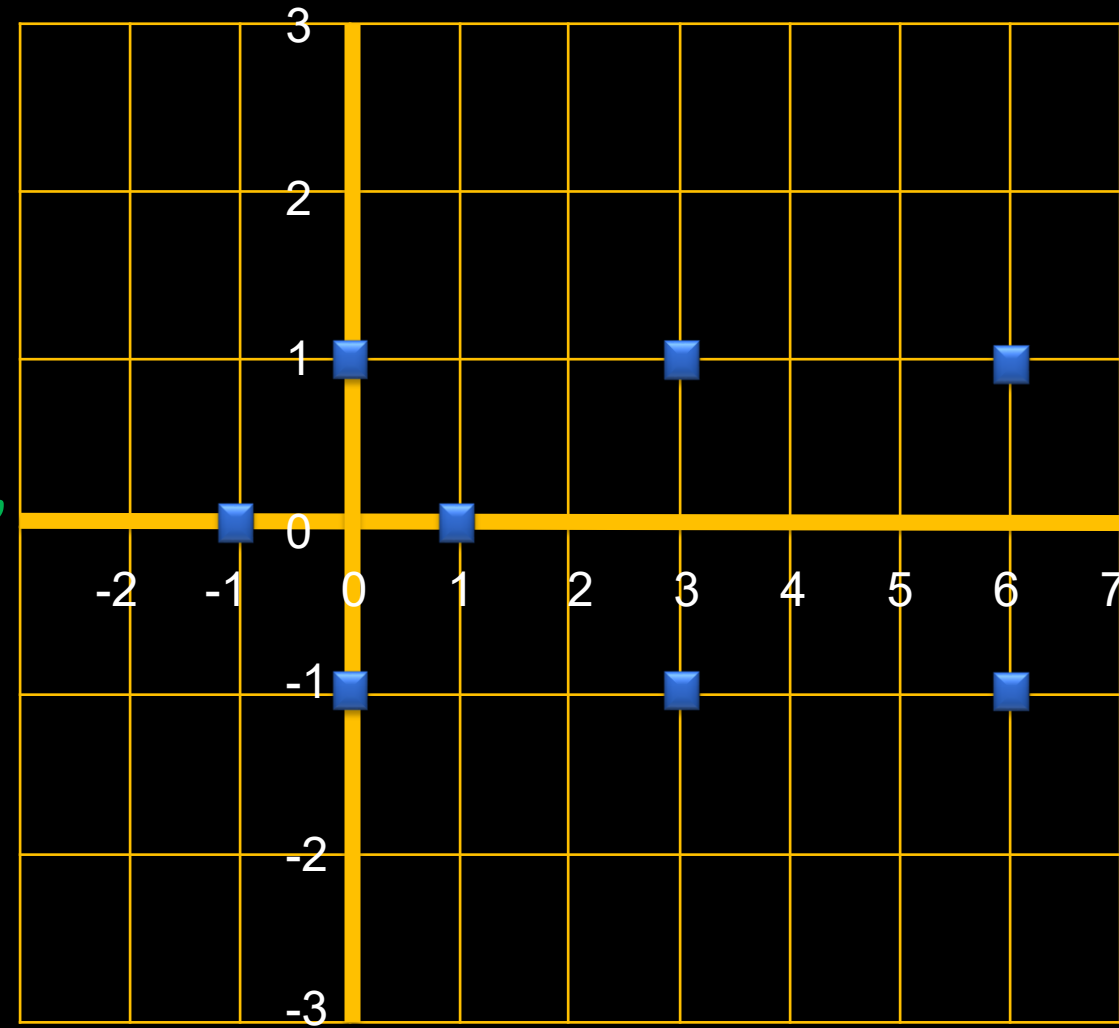
The objective is to find a **good balance** between keeping the street as large as possible and limiting the **margin violations**.



Linear SVM Example

- Suppose we are given the following data points

$\{(3,1), (3,-1), (6,1), (6,-1), (1,0),$
 $(0,1), (0,-1), (-1,0)\}$



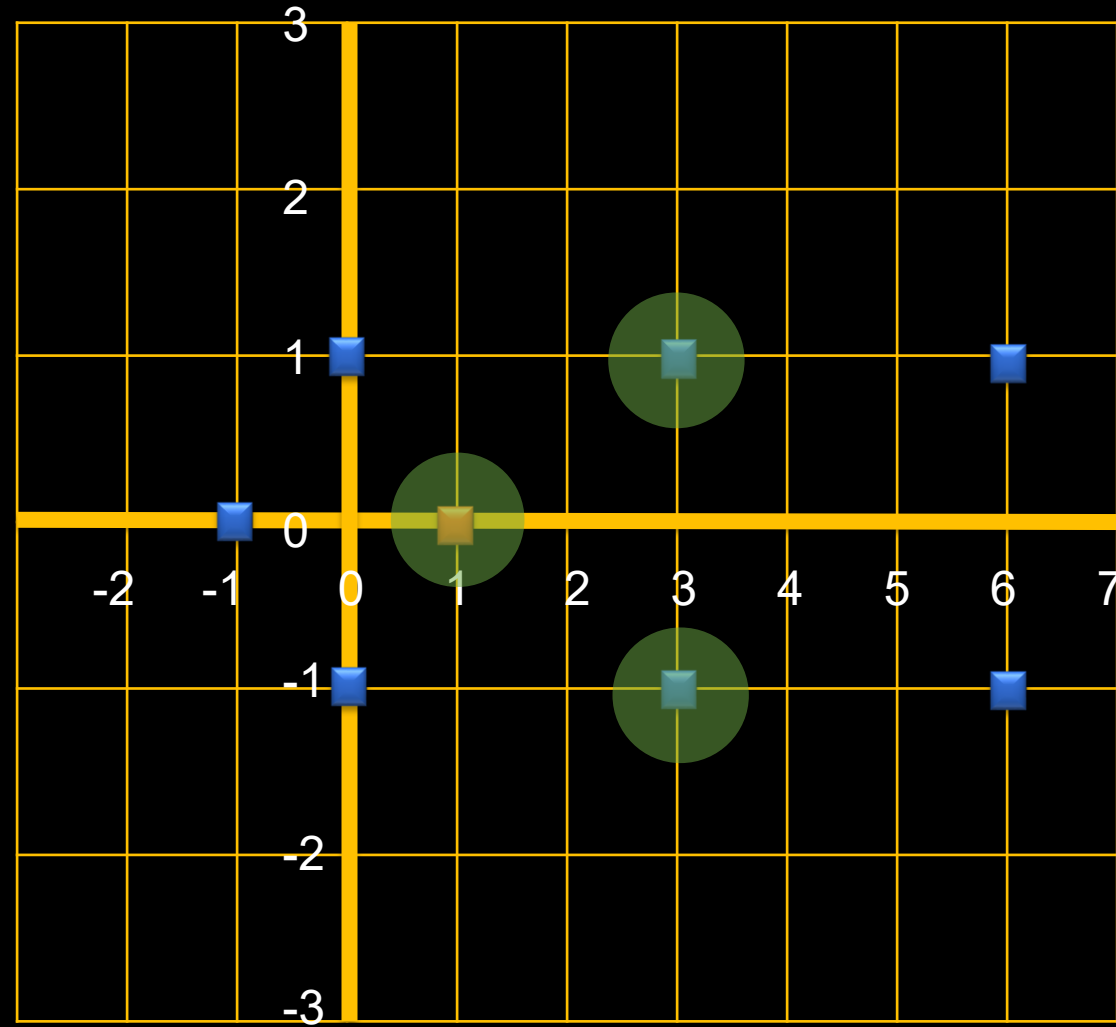
SVM Example

- By inspection, it should be obvious that there are **THREE** support vectors

$$s_1 = (1, 0)$$

$$s_2 = (3, 1)$$

$$s_3 = (3, -1)$$



SVM Example

- Each vector is augmented with a 1 as a **bias** input

$$\{s_1=(1,0), s_2=(3,1), s_3=(3,-1)\}$$

So,

$$s_1=(1,0), \text{ then } s_1^{\sim}=(1,0,1)$$

Similarly,

$$s_2=(3,1), \text{ then } s_2^{\sim}=(3,1,1)$$

$$s_3=(3,-1), \text{ then } s_3^{\sim}=(3,-1,1)$$

SVM Example

$$\alpha_1 s_1^{\sim} \cdot s_1^{\sim} + \alpha_2 s_2^{\sim} \cdot s_1^{\sim} + \alpha_3 s_3^{\sim} \cdot s_1^{\sim} = -1$$

$$\alpha_1 s_1^{\sim} \cdot s_2^{\sim} + \alpha_2 s_2^{\sim} \cdot s_2^{\sim} + \alpha_3 s_3^{\sim} \cdot s_2^{\sim} = +1$$

$$\alpha_1 s_1^{\sim} \cdot s_3^{\sim} + \alpha_2 s_2^{\sim} \cdot s_3^{\sim} + \alpha_3 s_3^{\sim} \cdot s_3^{\sim} = +1$$

Put the value of $s_1^{\sim}$, $s_2^{\sim}$ and $s_3^{\sim}$

$$\alpha_1(1,0,1) \cdot (1,0,1) + \alpha_2(3,1,1) \cdot (1,0,1) + \alpha_3(3,-1,1) \cdot (1,0,1) = -1$$

$$\alpha_1(1,0,1) \cdot (3,1,1) + \alpha_2(3,1,1) \cdot (3,1,1) + \alpha_3(3,-1,1) \cdot (3,1,1) = +1$$

$$\alpha_1(1,0,1) \cdot (3,-1,1) + \alpha_2(3,1,1) \cdot (3,-1,1) + \alpha_3(3,-1,1) \cdot (3,-1,1) = +1$$

SVM Example

$$\alpha_1(1,0,1) \cdot (1,0,1) + \alpha_2(3,1,1) \cdot (1,0,1) + \alpha_3(3,-1,1) \cdot (1,0,1) = -1$$

$$\alpha_1(1,0,1) \cdot (3,1,1) + \alpha_2(3,1,1) \cdot (3,1,1) + \alpha_3(3,-1,1) \cdot (3,1,1) = +1$$

$$\alpha_1(1,0,1) \cdot (3,-1,1) + \alpha_2(3,1,1) \cdot (3,-1,1) + \alpha_3(3,-1,1) \cdot (3,-1,1) = +1$$

Perform Dot Product

$$\alpha_1(1.1+0.0+1.1) + \alpha_2(3.1+1.0+1.1) + \alpha_3(3.1 + -1.0 + 1.1) = -1$$

$$\alpha_1(1.3+0.1+1.1) + \alpha_2(3.3+1.1+1.1) + \alpha_3(3.1 + -1.1 + 1.1) = +1$$

$$\alpha_1(1.3+0.-1+1.1) + \alpha_2(3.3+1.-1+1.1) + \alpha_3(3.3+-1.-1+1.1) = +1$$

SVM Example

$$\alpha_1(1.1+0.0+1.1) + \alpha_2(3.1+1.0+1.1) + \alpha_3(3.1 + -1.0 + 1.1) = -1$$

$$\alpha_1(1.3+0.1+1.1) + \alpha_2(3.3+1.1+1.1) + \alpha_3(3.1 + -1.1 + 1.1) = +1$$

$$\alpha_1(1.3+0.-1+1.1) + \alpha_2(3.3+1.-1+1.1) + \alpha_3(3.3+-1.-1+1.1) = +1$$

Simplify

$$\alpha_1(1 + 0 + 1) + \alpha_2(3 + 0 + 1) + \alpha_3(3 + 0 + 1) = -1 ; \quad 2\alpha_1 + 4\alpha_2 + 4\alpha_3 = -1$$

$$\alpha_1(3 + 0 + 1) + \alpha_2(9 + 1 + 1) + \alpha_3(3 - 1 + 1) = +1 ; \quad 4\alpha_1 + 11\alpha_2 + 9\alpha_3 = 1$$

$$\alpha_1(3 + 0 + 1) + \alpha_2(9 - 1 + 1) + \alpha_3(9 + 1 + 1) = +1 ; \quad 4\alpha_1 + 9\alpha_2 + 11\alpha_3 = 1$$

SVM Example

$$4\alpha_1 + 20\alpha_3 = 1 \text{ Put } \alpha_3 = 3/4$$

$$4\alpha_1 + 20 \times 3 / 4 = 1$$

$$4\alpha_1 + 5 \times 3 = 1$$

$$4\alpha_1 = -14 ; \alpha_1 = -14 / 4 = -3.5$$

$$\alpha_1 = -3.5$$

$$\alpha_2 = 0.75$$

$$\alpha_3 = 0.75$$

$$\alpha_1(1 + 0 + 1) + \alpha_2(3 + 0 + 1) + \alpha_3(3 + 0 + 1) = -1 ; \quad 2\alpha_1 + 4\alpha_2 + 4\alpha_3 = -1$$

$$\alpha_1(3 + 0 + 1) + \alpha_2(9 + 1 + 1) + \alpha_3(3 - 1 + 1) = +1 ; \quad 4\alpha_1 + 11\alpha_2 + 9\alpha_3 = 1$$

$$\alpha_1(3 + 0 + 1) + \alpha_2(9 - 1 + 1) + \alpha_3(9 + 1 + 1) = +1 ; \quad 4\alpha_1 + 9\alpha_2 + 11\alpha_3 = 1$$

Simplify

$$4\alpha_1 + 9\alpha_2 + 11\alpha_3 = 1$$

$$4\alpha_1 + 11\alpha_2 + 9\alpha_3 = 1$$

$$\begin{array}{r} - \\ - \\ - \\ - \\ \hline -2\alpha_2 + 2\alpha_3 = 0 \end{array}$$

$$\alpha_2 = \alpha_3$$

$$2(2\alpha_1 + 4\alpha_2 + 4\alpha_3 = -1) \text{ Multiply both side by 2}$$

$$4\alpha_1 + 8\alpha_2 + 8\alpha_3 = -2 \text{ Replace } \alpha_2 \text{ with } \alpha_3$$

$$4\alpha_1 + 8\alpha_3 + 8\alpha_3 = -2$$

$$4\alpha_1 + 16\alpha_3 = -2$$

$$4\alpha_1 + 9\alpha_2 + 11\alpha_3 = 1$$

$$4\alpha_1 + 9\alpha_3 + 11\alpha_3 = 1$$

$$4\alpha_1 + 20\alpha_3 = 1$$

$$4\alpha_1 + 16\alpha_3 = -2$$

$$4\alpha_1 + 20\alpha_3 = 1$$

$$\begin{array}{r} - \\ - \\ - \\ \hline -4\alpha_3 = -3 \end{array}$$

$$\alpha_3 = 3 / 4 = 0.75$$

SVM Example

$$\bar{w} = \sum \alpha_i s_i$$

$$= -3.5 (1,0,1) + 0.75(3,1,1) + 0.75 (3,-1,1)$$

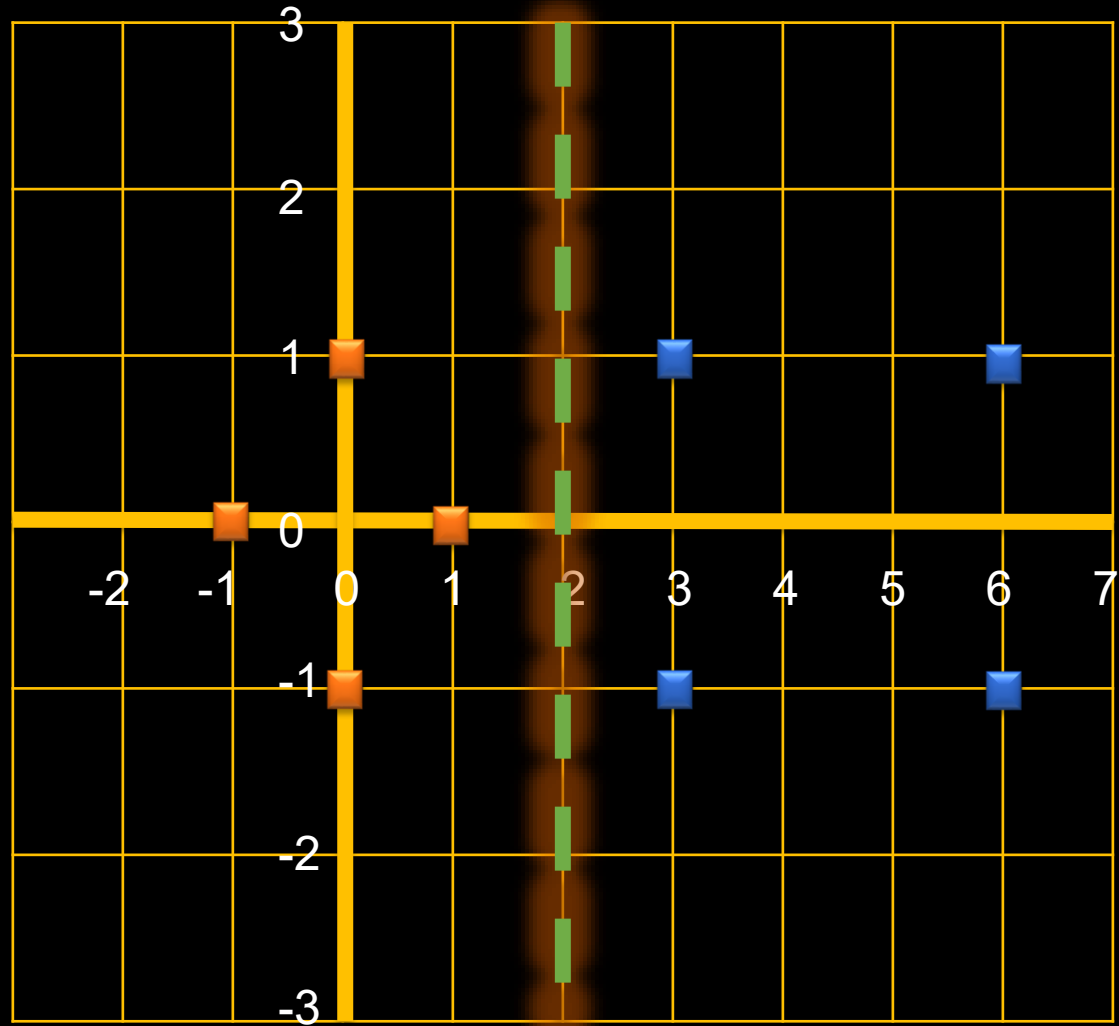
$$= (-3.5, 0, -3.5) + (2.25, 0.75, 0.75) + (2.25, -0.75, 0.75)$$

$$= (-3.5 + 2.25 + 2.25, 0 + 0.75 - 0.75, -3.5 + 0.75 + 0.75)$$

$$= (1, 0, -2)$$

Finally, **Hyperplane** equation $y = w x + b$ where $w = (1, 0)$ & $b = -2$

SVM Example

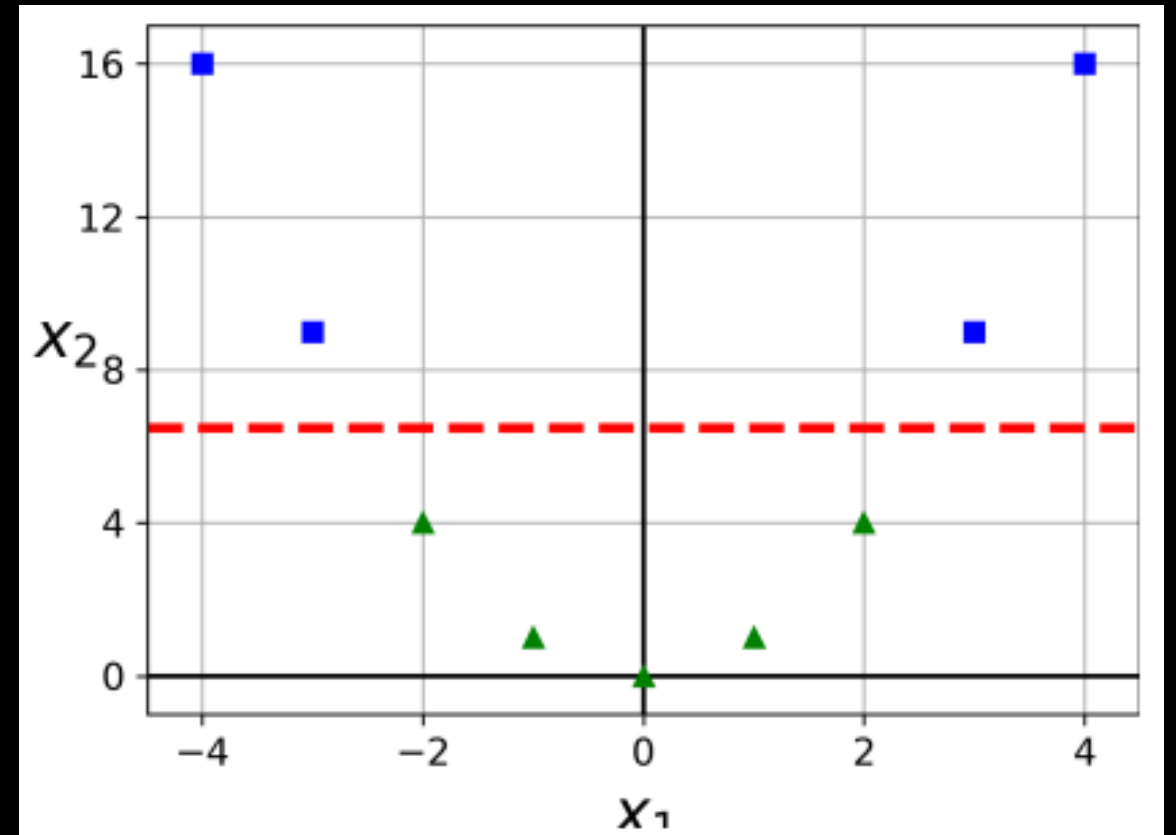
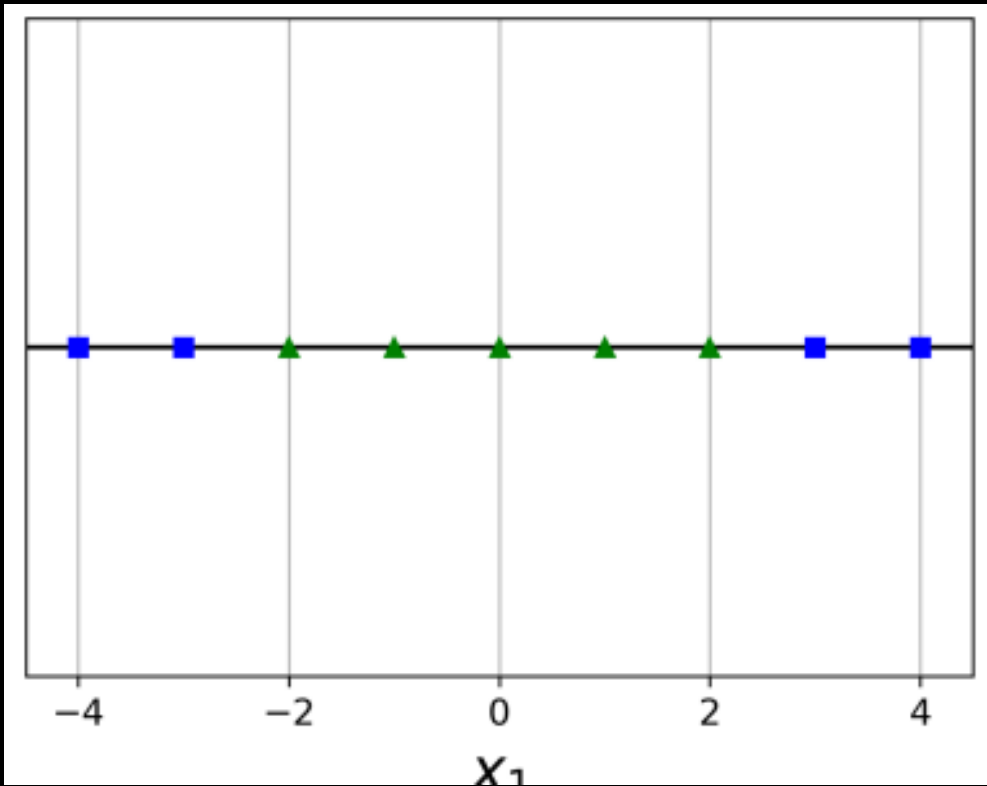


Finally, **Hyperplane** equation $y = w x + b$ where $w = (1, 0)$ & $b = -2$

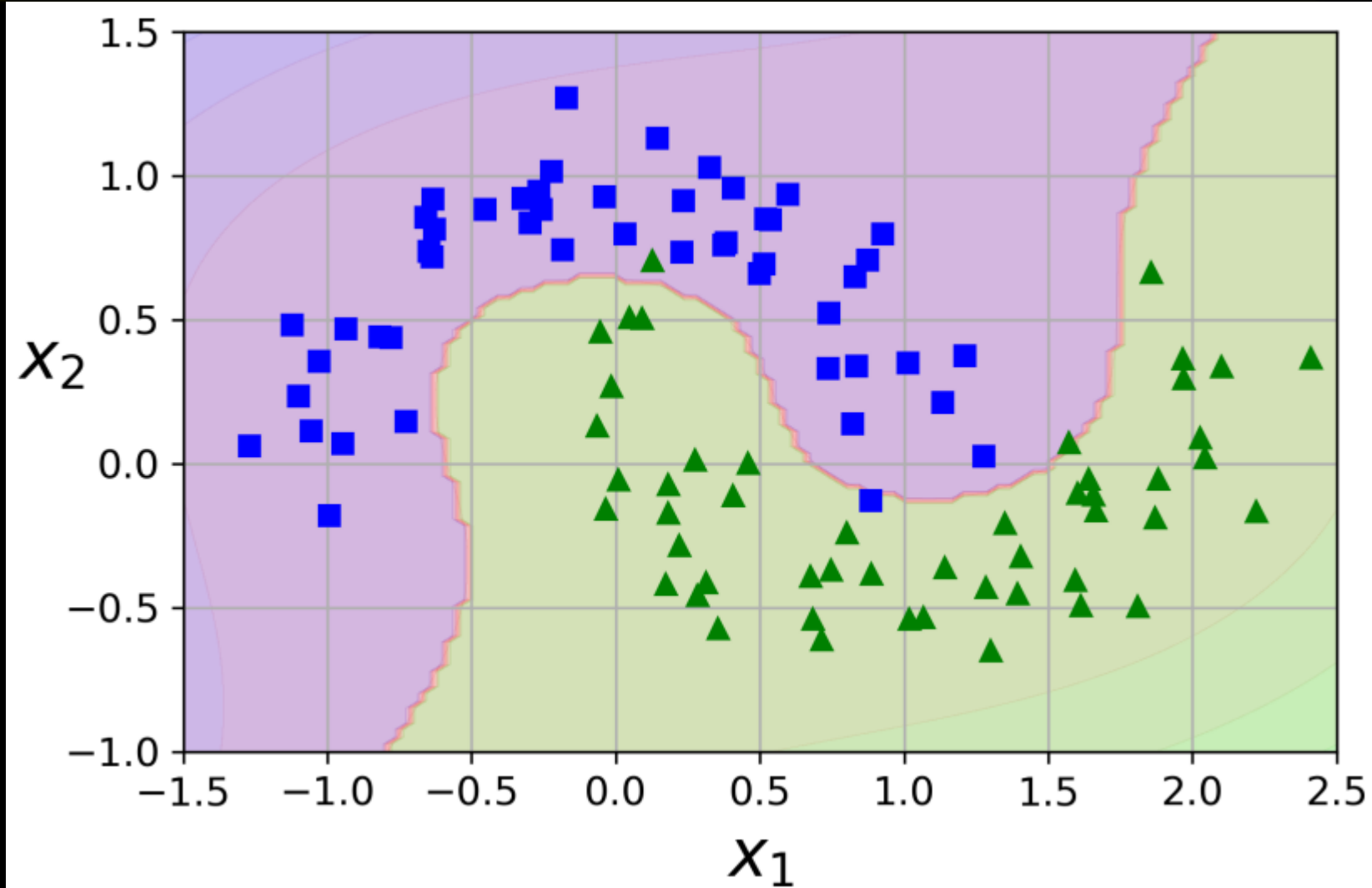
Non-Linear SVM (Support Vector Machine)

- **Linear SVM classifiers** are **efficient** and work **surprisingly** well in many cases, many datasets are not even close to being linearly separable.
- One approach to handling **nonlinear datasets** is to **add more features**, such as **polynomial features**.

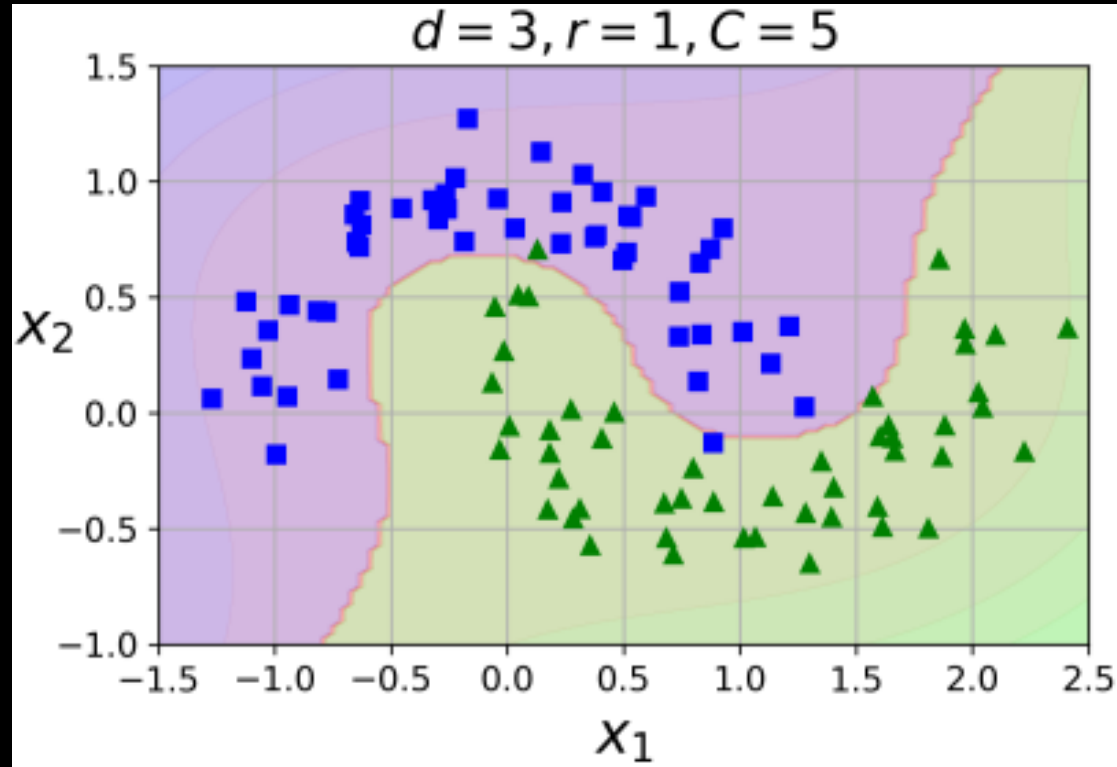
Non-Linear SVM (Support Vector Machine)



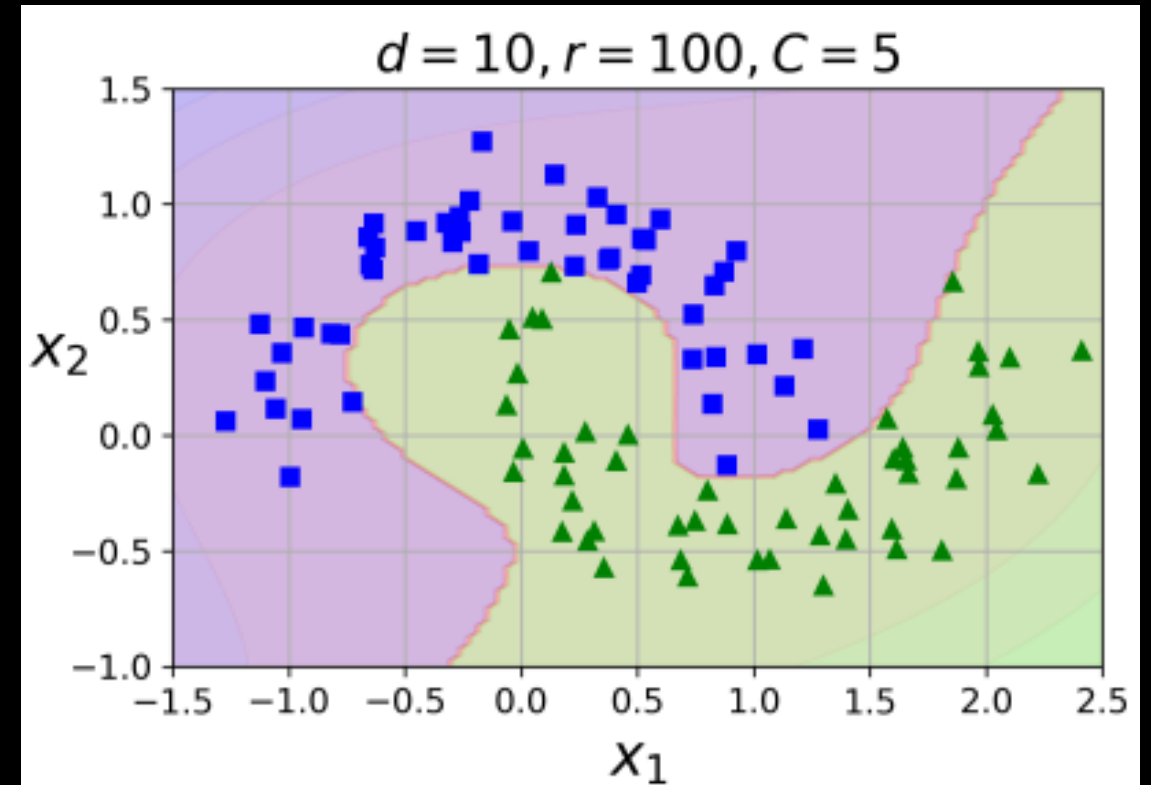
Polynomial Kernel (Non-Linear SVM)



Polynomial Kernel (Non-Linear SVM)



3rd-degree Polynomial Kernel



Non Linear SVM Example

- Suppose we are given the following positively data points

$\{(2,2), (2,-2), (-2,-2), (-2,2)\}$

- And the following negatively labeled data points,

$\{(1,1), (1,-1), (-1,-1), (-1,1)\}$

Non Linear SVM Example

- Convert data from one feature space to another.

$$A_1(x_1, x_2) = \begin{cases} (4 - x_2 + |x_1 - x_2|) & \text{If } \sqrt{x_1^2 + x_2^2} > 2 \\ (4 - x_1 + |x_1 - x_2|) & \\ (x_1, x_2) & \text{Otherwise} \end{cases}$$

Non Linear SVM Example

- Positive Examples

$$s_1=(2,2), s_2=(2,-2), s_3=(-2,-2), s_4=(-2,2)$$

Take s_1 First check the condition If $S_1^2 + S_2^2 > 2$

$$s_1=(2,2), s_2=(10,6), s_3=(6,6), s_4=(6,10)$$

- Negative Examples

$$s_1=(1,1), s_2=(1,-1), s_3=(-1,-1), s_4=(-1,1)$$

$$s_1=(1,1), s_2=(1,-1), s_3=(-1,-1), s_4=(-1,1)$$

Non Linear SVM Example

- Student Task

Non Linear SVM Example

$$\vec{w} = \sum \alpha_i s_i$$

$$= -7(1,1,1) + 4(2,2,1)$$

$$= (-7,-7,-7) + (8,8,4)$$

$$= (-7 + 8, -7 + 8, -7 + 4)$$

$$= (1, 1, -3)$$

Finally, **Hyperplane** equation $y = w x + b$ where $w = (1, 1)$ & $b = -3$

Non Linear SVM Example

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